Advanced Structural Analysis using ANSYS

Lecture and Exercise Notes

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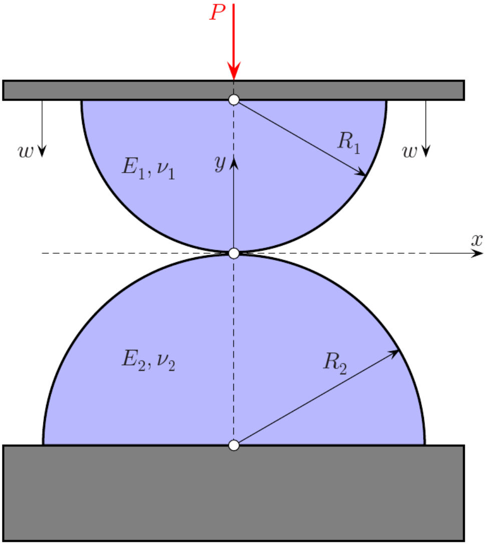
Exercise 6: Contact between elastic semi-cylinder

Die Veranstaltung basiert u.a. auf dem Vorlesungs- und Übungsskript „Die Methode der finiten Elemente“, gehalten durch Prof. Dr. rer. nat. M. Braun, Dr.-Ing. H. Walter, Dr.-Ing. U. Zwiers und Dr.-Ing. Philipp Olle, am Lehrstuhl für Mechanik (Maschinenbau) der Universität Duisburg-Essen im WS2004/05

Contact between elastic semi-cylinder

Problem description



- Geometric data
  $R_1 = 100\,\mathrm{mm}$
  $R_2 = 120\,\mathrm{mm}$
- Material data
  Box (Aluminum):
  $E_1 = 180{,}000\,\mathrm{N/mm^2}$
  $E_2 = 210{,}000\,\mathrm{N/mm^2}$
  $\nu_1 = 0.28$
  $\nu_2 = 0.3$

  Both contact surfaces are not rigid and frictionless
- Load

  $P = 2000\,\mathrm{N/mm}$
- Objective target

  (final) displacement w
  semi contact width a
  maximum pressure p_0

Frictionless contact between two elastic cylinders

reference solution

references:

H. Hertz: *Über die Berührung fester elastischer Körper*. Journal für die reine und angewandte Mathematik (1882).

K.L. Johnson: *Contact Mechanics*. Cambridge University Press, Cambridge (1985).

half contact width (Hertz) $$a = \sqrt{\frac{4PR}{\pi E}}$$

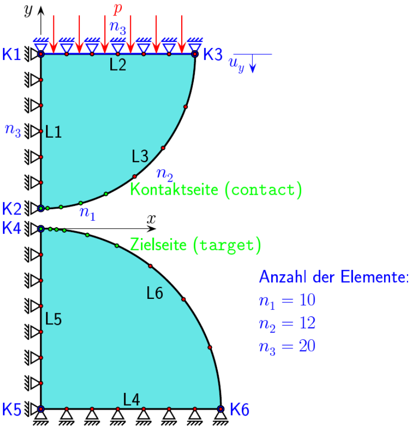
max surface pressure (Hertz) $$p_0 = \sqrt{\frac{PE}{\pi R}}$$

with $$E = \left(\frac{1-\nu_1^2}{E_1} + \frac{1-\nu_2^2}{E_2}\right)^{-1}, \quad R = \left(\frac{1}{R_1} + \frac{1}{R_2}\right)^{-1}$$

displacement (Johnson) $$w = \frac{2P}{\pi}\left[\frac{1-\nu_1^2}{E_1}\ln\frac{4R_1}{a} + \frac{1-\nu_2^2}{E_2}\ln\frac{4R_2}{a} - \frac{1}{2E}\right]$$

3D model, boundary condition, load

Contact between elastic semi-cylinder



- Element types
  - 2-D PLANE182
    (Simplified enhanced strain formulation)
    (plane-strain-condition)

- Boundary conditions

Nodes	$u_x = 0$	$u_y = 0$	coupling u_y
$x = 0$	×		
$y = -R_2$		×	
$y = R_1 +$ inital gap			×

- Meshing and Load

  free meshing, segmentation of arcs $3:7$, max mesh refinement at keypoint 2 & 4, contact modeling using *Contact Wizard*, load as pressure on upper line
  inital gap $= 10\,\mathrm{mm}$

- Load steps

  4 load steps, see next slide

Load stepping

Contact between elastic semi-cylinder

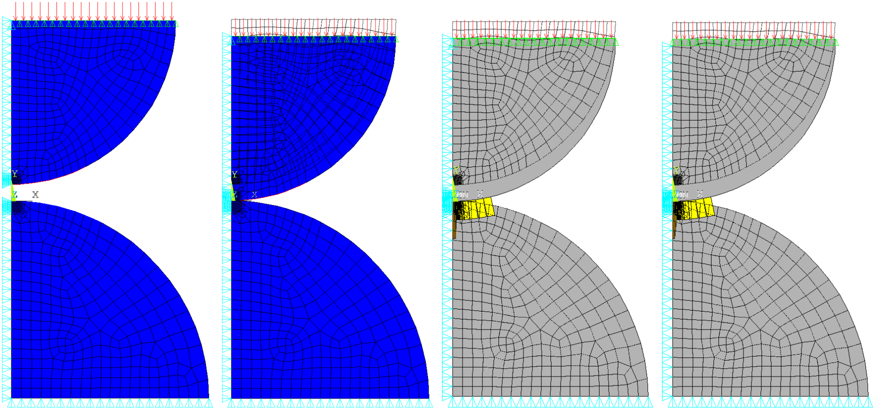
Load steps

1. fix vertical displacement of upper line ($y = R_1 +$ initial gap)
   apply pressure on this line
   $\rightarrow$ sub-stepping: 10 steps, max 20, min 5
2. close contact by applying displacement on upper line
   $\rightarrow$ sub-stepping: 10 steps, max 20, min 5
3. apply additional displacement ($u = 0.1$) on upper line to assign a static contact situation
   $\rightarrow$ sub-stepping: 20 steps, max 100, min 10
4. remove displacement boundary condition on upper line, thus, only pressure will "work"
   $\rightarrow$ sub-stepping: 100 steps, max 1000, min 50

Load stepping (cntd.)

Contact between elastic semi-cylinder

1st LS | 2nd LS | 3rd LS | 4th LS



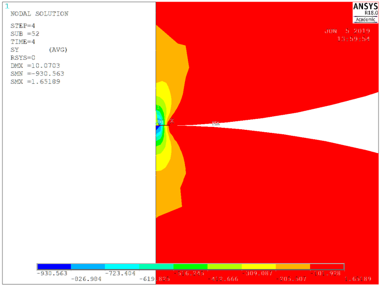
$u_y = -0\,\mathrm{mm}$ | $u_y = -10.0\,\mathrm{mm}$ | $u_y = -10.1\,\mathrm{mm}$ | u_y is only coupled, not Dirichlet bounded

Post processing

Contact between elastic semi-cylinder

- final displacement w of upper line due to pressure P
  $w = -10.070\,\mathrm{mm}$
- maximum contact pressure
  $p_0 = -963.14\,\mathrm{N/mm^2}$
- semi contact width a
  1. get nodes with contact pressure larger than zero
  2. among those, get the node with the maximum x-coordinate $\rightarrow x_{max}$
  3. get the x-displacement of this node $\rightarrow u_x$
  4. $a = x_{max} + u_x = 1.5706\,\mathrm{mm} + (-0.002948\,\mathrm{mm}) = 1.5677\,\mathrm{mm}$

stresses σ_{22}



contact pressure